Problem · Topic 3 · 4 points · 60 min

P11 — FRIGOLEV: an incomplete 2017 balance, a cooperative discount and the 2018 statements

The newly appointed financial director of ***LOGÍSTICA FRIGORÍFICA DEL LEVANTE S.L. (FRIGOLEV)****, a company located in the port area of Castellón and dedicated to the manufacturing of isothermal containers and cold-chain storage equipment for agricultural exporters, is working on the closing of its financial statements (balance sheet and Income Statement) as of 31 December 2018 (31/12/2018).*

The company was originally incorporated on 1 January 2016 (01/01/2016) by four partners, with an initial contribution of each of 1.500 m.u. disbursed fully in cash at that time, and no further capital modifications took place until fiscal year 2018. Due to the sudden departure of the previous chief accountant, the financial director only has an incomplete balance of the company as of 31 December 2017 (31/12/2017), which is included below with certain missing figures marked as (?), along with the audit notes detailing the operations that have taken place during the period from 1 January 2018 (01/01/2018) to 31 December 2018 (31/12/2018):

I. *On 31 December 2017 (31/12/2017), the company had as depreciable fixed assets only the industrial refrigeration chambers purchased on a certain date during 2016, which have a useful life of 10 years and are amortized linearly with a residual value of 10% of their acquisition cost. In addition, the balance sheet at 31 December 2017 (31/12/2017) reflected an industrial plot of land acquired in 2016 and financial shares of other companies acquired as a permanent investment.*

II. *During the year 2018, the company has sold 3.600 units of isothermal containers at a catalog price of 25 m.u. / unit (every month the same amount, 300 u.). The general payment condition agreed with the customers implies that this will occur 60 days after the sale. However, a very important agricultural cooperative (representing 25% of the physical units sold each month), which has bought an equal amount every month, has been given a discount of 8% on the catalog price of their purchases in exchange for paying strictly in cash in the same month of the sale.*

III. *At 31 December 2017 (31/12/2017) there was a stock (stock in stock) of 400 tons of insulating raw material valued at 2,00 m.u. / ton. On 1 April 2018 (01/04/2018), the company purchased 5.000 tons of raw material at a cost of 2,00 m.u. / ton. The payment condition agreed with the suppliers was 40% paid in cash on the date of purchase, 30% payable to 90 days, and the remaining 30% deferred until 1 February 2019 (01/02/2019). At 31 December 2018 (31/12/2018), after conducting the warehouse count, there remains a final stock (stock in stock) of 900 tons.*

IV. *During the first quarter of 2018, the unpaid taxes and the outstanding supplier balances from the previous year were paid in full. Regarding the customer debt outstanding from 31 December 2017 (31/12/2017), 75% was charged in cash during the first semester of 2018, except 200 m.u. from a customer who requested and was granted a deferral of payment until 15 March 2019 (15/03/2019), and 100 m.u. corresponding to a client declared definitively insolvent on 1 June 2018 (01/06/2018), which must be recorded as an unrecoverable bad debt loss for the period.*

V. *Fixed personnel expenses have been 2.200 m.u. / month (to which 600 m.u. / month must be added for Social Security) from 1 January 2018 (01/01/2018). To respond to seasonal peaks in container assembly, production staff also received variable wages that amounted in 2018 to 3.620 m.u. Fixed and variable wages are paid in the month in which they are produced, whereas Social Security contributions (starting from January 2018) are paid on the 15th day of the following month. General expenses and utilities (electricity, phone, insurance, etc.) amounted to 1.100 m.u. / month and are paid two months after being made.*

VI. *The company carried out a renting contract on 1 March 2018 (01/03/2018) (with a duration of 2 years) with a specialized company for electric warehouse forklifts, for a value of 350 m.u. monthly. The contract establishes a condition of payment to 90 days (payment at the end of each month).*

VII. *On 1 April 2018 (01/04/2018), the company made a capital increase by issuing 1.000 new shares of 2,00 m.u. nominal value each, fully disbursed in cash and with an issue premium of 50% over the nominal value. On the same date, 1 April 2018 (01/04/2018), it bought two refrigerated delivery vehicles that cost a total of 4.800 m.u., of which it paid 50% in cash, postponing the payment of the remainder in two equal installments: half on 1 October 2018 (01/10/2018) and the other half on 1 April 2019 (01/04/2019). These vehicles will be amortized linearly in 6 years with a total residual value of 480 m.u. Furthermore, on 1 October 2018 (01/10/2018), it bought a software license for cold-chain tracking for a value of 1.600 m.u., payable to 120 days, which is amortized linearly in 4 years with residual value zero.*

VIII. *The long-term bank loan appearing on the balance sheet at 31 December 2017 (31/12/2017) was granted on 1 January 2017 (01/01/2017) at an annual interest rate of 6%. Each 31 December, the company must pay the accrued annual interest calculated from the loan value at the beginning of each year, and starting on 31 December 2018 (31/12/2018) it must also return to the bank an annual principal amount of 500 m.u. until completing the repayment on 31 December 2021 (31/12/2021). Additionally, on 1 July 2018 (01/07/2018), the company obtained a new short-term bank credit of 1.500 m.u. at 8% annual interest, agreeing to return 50% of the principal each semester for the next year. Both the principal installment and the semiannual interest calculated on the amount actually due in that semester will be paid the day after the end of each semester (i.e., on 1 January 2019 (01/01/2019) and 1 July 2019 (01/07/2019)).*

IX. *The financial results of the company are declared at 23:59 hrs on 31 December 2018 (31/12/2018). The applicable corporation tax rate is 25% on the Profits obtained, which is settled by calendar year and will be paid to the Treasury on 20 January 2019 (20/01/2019) (counted in the balance at 31 December 2018 (31/12/2018) as "unpaid taxes"). From the net profit after taxes, the company first offsets in full the losses from previous years, and of the remaining net profit, it retains 60% as reserves and distributes the other 40% as dividends to the shareholders, which are paid in cash before carrying out the final balance at 24:00 on 31 December 2018 (31/12/2018).*

BALANCE 31 December 2017 (31/12/2017)

ASSETS	*m.u.*	*LIABILITIES*	*m.u.*
CURRENT ASSETS	***2.940***	***CURRENT LIABILITIES***	***900***
Cash	*(?)*	*Unpaid taxes*	*250*
Accounts receivable	*1.200*	*Accounts payable (suppliers)*	*650*

ASSETS	*m.u.*	*LIABILITIES*	*m.u.*
Inventory (stock)	*800*	***NON-CURRENT LIABILITIES***	***2.000***
NON-CURRENT ASSETS	***5.560***	*Long-term bank loan*	*2.000*
Land	*1.500*	***OWNERS EQUITY***	***(?)***
Refrigeration chambers (Equipment)	*4.000*	*Paid-in capital*	*(?)*
Accum. depreciation	*(540)*	*Losses from previous years*	*(400)*
Financial shares	*600*		
Total Assets	***(?)***	***Total Liabilities + OE***	***(?)***

Required:

1. *Determine and justify the missing figures (?) in the balance sheet as of 31 December 2017 (31/12/2017) as well as the exact date in 2016 on which the industrial refrigeration chambers were purchased, and construct the Income Statement (Profit and Loss Account) of FRIGOLEV for the period ending 31 December 2018 (31/12/2018).*
2. *Assuming that the company has offset the prior losses and paid the dividends as indicated, construct and justify the balance sheet at 24:00 on 31 December 2018 (31/12/2018), detailing all intermediate calculations.*
3. *Determine the values of the ROA and ROE ratios in 2018, and indicate which items of the balance sheet at 24:00 on 31 December 2018 (31/12/2018) would be affected, and the exact amount of the effect, if the decision was made before the end of 2018 to distribute the same dividend, but to pay it on 15 April 2019 (15/04/2019).*
4. *State and explain what changes would occur on the various items of the Income Statement and the balance sheet at 31 December 2018 (31/12/2018) if the company had eliminated the 8% discount granted to the agricultural cooperative, subjecting this customer to the general 60-day payment condition at the full price of 25 m.u. / unit, which in turn would have caused the cooperative to reduce the physical units bought each month by 10%. In which case is the net profit higher, and in which case is the final cash balance higher? Discuss reasoned, why.*

Setup

Hypotheses (readings of the statement that the solution depends on):

1. Item IV: 75% of the 1.200 m.u. (900 m.u.) is collected; the remaining 25% (300 m.u.) is exactly the 200 m.u. deferred plus the 100 m.u. written off.
2. Item IX: the 25% tax is applied to the whole profit before taxes of 2018; the losses of previous years are offset afterwards, against the net profit, as the statement orders.
3. Item VI: "payable to 90 days (payment at the end of each month)" means each month's fee is paid at the end of the third month after it (March is paid at the end of June); October, November and December are unpaid at 31/12/2018.
4. Requirement 4: the containers consume raw material in proportion to the units sold, and no finished containers are kept in stock.

INCONSISTENCY IN THE GIVEN BALANCE

Item VIII says that 500 m.u. of the loan are repaid on 31/12/2018, so at 31/12/2017 those 500 m.u. were due within one year and should have been a current liability (current liabilities 1.400, non-current 1.500). The totals do not change, so the solution keeps the given balance as stated and splits the loan correctly at 31/12/2018.

Timeline of the cash-relevant dates in 2018

Date	Event
Q1 2018	Unpaid taxes 250 and suppliers 650 of 2017 paid
H1 2018	900 of the 2017 receivables collected; 100 written off on 01/06
01/04/2018	Raw material 10.000 (4.000 cash); capital increase 3.000; vehicles 4.800 (2.400 cash)
01/07/2018	Second 30% of raw material paid (3.000); short-term credit received 1.500
01/10/2018	Second vehicle instalment 1.200; software 1.600 bought payable to 120 days
31/12/2018	Loan interest 120 and principal 500 paid; tax accrued; dividends paid at 24:00

Extension (not defined in the course decks). $\mathrm{ROA} = \text{Income from operations}/\text{Total assets}$ and $\mathrm{ROE} = \text{Net income}/\text{Owners equity}$, both on the balance at 24:00 on 31/12/2018, as in the solved INFOPYMES exercise.

Part (a) · 1.25 points

Find the missing figures of the 2017 balance and the purchase date of the chambers, and prepare the 2018 Income Statement

Step 1. The missing figures of 31/12/2017

$$\text{Cash} = 2.940 - 1.200 - 800 = 940 \text{ m.u.}, \qquad \text{Total assets} = 2.940 + 5.560 = 8.500 \text{ m.u.}$$

Check of the non-current block: $1.500 + 4.000 - 540 + 600 = 5.560$ (checked).

$$\text{Paid-in capital} = 4 \times 1.500 = 6.000, \qquad \text{Owners equity} = 6.000 - 400 = 5.600 \text{ m.u.}$$

$$\text{Total liabilities} + \text{OE} = 900 + 2.000 + 5.600 = 8.500 \text{ m.u.} \checkmark$$

Step 2. Purchase date of the refrigeration chambers

$$D_{\text{year}} = \frac{4.000 - 0{,}10 \times 4.000}{10} = \frac{3.600}{10} = 360 \text{ m.u./year}$$

$$\frac{540}{360} = 1{,}5 \text{ years before } 31/12/2017 \;\Rightarrow\; \text{bought on 1 July 2016 } (01/07/2016)$$

Step 3. Revenue and its collection (item II)

Each month 300 units are sold: 75 to the cooperative (25%) at $25 \times 0{,}92 = 23$ m.u. and 225 to the other customers at 25 m.u.

$$\text{Cooperative} = 75 \times 23 = 1.725 \text{ m.u./month}, \qquad \text{Others} = 225 \times 25 = 5.625 \text{ m.u./month}$$

$$\text{Revenue 2018} = 12 \times (1.725 + 5.625) = 12 \times 7.350 = 88.200 \text{ m.u.}$$

The cooperative pays in the same month (20.700 collected); the others pay at 60 days, so November and December are pending: $2 \times 5.625 = 11.250$ m.u. receivable.

Step 4. Raw material (item III)

$$\text{Consumption} = 800 + 5.000 \times 2{,}00 - 900 \times 2{,}00 = 800 + 10.000 - 1.800 = 9.000 \text{ m.u.}$$

Payments: 40% on 01/04 (4.000) and 30% at 90 days, on 01/07 (3.000); the last 30% (3.000) is owed until 01/02/2019.

Step 5. Personnel, general expenses and renting (items V, VI)

$$\text{Personnel} = 12 \times 2.200 + 12 \times 600 + 3.620 = 26.400 + 7.200 + 3.620 = 37.220 \text{ m.u.}$$

- Social Security of December (600) is paid on 15/01/2019; 11 months paid (6.600).

- General expenses: $12 \times 1.100 = 13.200$ m.u. of expense; paid two months later, so November and December (2.200) are owed and 11.000 paid.
- Renting: March to December, $10 \times 350 = 3.500$ m.u. of expense; March to September paid ($7 \times 350 = 2.450$), October to December owed (1.050).

Step 6. Depreciation and amortization (items I, VII)

$$D_{\text{chambers}} = 360, \qquad D_{\text{vehicles}} = \frac{4.800 - 480}{6} \times \frac{9}{12} = 720 \times \frac{9}{12} = 540, \qquad D_{\text{software}} = \frac{1.600}{4} \times \frac{3}{12} = 100$$

$$D_{2018} = 360 + 540 + 100 = 1.000 \text{ m.u.}$$

Step 7. Financial expenses (item VIII)

$$I_{\text{loan}} = 0{,}06 \times 2.000 = 120 \text{ m.u. (paid } 31/12/2018), \qquad I_{\text{credit}} = 1.500 \times 0{,}08 \times \frac{6}{12} = 60 \text{ m.u. (paid } 01/01/2019)$$

Step 8. Income Statement 2018

Line	Computation	m.u.
Sales revenue	Step 3	88.200
– Raw material consumed	Step 4	9.000
– Personnel (wages and Social Security)	Step 5	37.220
– General expenses and utilities	12×1.100	13.200
– Renting of forklifts	10×350	3.500
– Bad debt loss	item IV	100
– Depreciation and amortization	Step 6	1.000
= Income from operations	$88.200 - 64.020$	**24.180**
– Financial expenses	$120 + 60$	180
= Profit before taxes	$24.180 - 180$	**24.000**
– Corporation tax (25%)	$0{,}25 \times 24.000$	6.000
= Net profit	$24.000 - 6.000$	**18.000**

KEY RESULTS

- Cash 940; total 8.500; paid-in capital 6.000; owners equity 5.600 m.u.
- Chambers depreciate 360 m.u./year: bought on 1 July 2016 (01/07/2016).
- Revenue 88.200; raw material consumed 9.000; depreciation 1.000 m.u.
- Income from operations 24.180; financial expenses 180; net profit 18.000 m.u.

Part (b) · 1.25 points

Prepare the balance sheet at 24:00 on 31 December 2018

Step 1. Distribution of the profit [Slide 23]

$$\text{Remaining profit} = 18.000 - 400 = 17.600, \qquad \text{Reserves} = 0{,}60 \times 17.600 = 10.560, \qquad \text{Dividends} = 0{,}40 \times 17.600 = 7.040 \text{ m.u.}$$

Step 2. Capital increase (item VII)

$$1.000 \times 2{,}00 = 2.000 \text{ (capital)}, \qquad 0{,}50 \times 2.000 = 1.000 \text{ (issue premium)}, \qquad \text{cash in} = 3.000 \text{ m.u.}$$

Step 3. Cash account 2018

Inflows	m.u.	Outflows	m.u.
Cooperative sales (cash)	20.700	Unpaid taxes 2017	250
Other customers, January–October (10×5.625)	56.250	Suppliers 2017	650
Receivables of 2017 (75%)	900	Raw material ($4.000 + 3.000$)	7.000
Capital increase	3.000	Fixed wages	26.400
Short-term bank credit	1.500	Variable wages	3.620
		Social Security (11 months)	6.600
		General expenses (10 months)	11.000
		Renting (7 months)	2.450
		Vehicles ($2.400 + 1.200$)	3.600
		Loan interest	120
		Loan principal	500
		Dividends	7.040
Total	**82.350**	**Total**	**69.230**

$$\text{Cash}_{24:00,\,31/12/2018} = 940 + 82.350 - 69.230 = 14.060 \text{ m.u.}$$

Step 4. Balance sheet at 24:00 on 31/12/2018

Assets	**m.u.**	**Liabilities and owners equity**	**m.u.**
Land	1.500	Paid-in capital (6.000 + 2.000)	8.000
Refrigeration chambers (4.000 − 900)	3.100	Issue premium	1.000
Vehicles (4.800 − 540)	4.260	Reserves	10.560
Software license (1.600 − 100)	1.500	Losses from previous years (offset)	0
Financial shares	600	**Owners equity**	**19.560**
Non-current assets	**10.960**	Long-term bank loan (2020–2021)	1.000
Inventory (900 t × 2,00)	1.800	**Non-current liabilities**	**1.000**
Accounts receivable (11.250 + 200)	11.450	Bank loan, instalment 31/12/2019	500
Cash	14.060	Short-term bank credit	1.500
		Interest payable (credit)	60
		Suppliers of raw material	3.000
		Suppliers of vehicles	1.200
		Supplier of software	1.600
		Social Security payable	600
		General expenses payable	2.200
		Renting payable	1.050
		Unpaid taxes	6.000
Current assets	**27.310**	**Current liabilities**	**17.710**
Total assets	**38.270**	**Total liabilities and equity**	**38.270**

Accumulated depreciation of the chambers: $540 + 360 = 900$. The 200 m.u. deferred to 15/03/2019 is due within one year, so it stays current; the loan splits into the 500 due on 31/12/2019 (current) and 1.000 due in 2020–2021.

KEY RESULTS

- Remaining profit 17.600; reserves 10.560; dividends 7.040 m.u.
- Cash at 24:00 on 31/12/2018: 14.060 m.u.
- Owners equity 19.560; current liabilities 17.710; total 38.270 m.u.

Part (c) · 0.75 points

Compute ROA and ROE of 2018, and the effect of paying the same dividend on 15 April 2019

Step 1. Ratios (extension)

$$\text{ROA} = \frac{24.180}{38.270} = 63{,}18\%, \qquad \text{ROE} = \frac{18.000}{19.560} = 92{,}02\%$$

Step 2. Dividend decided in 2018, paid on 15/04/2019

Item	Paid at 24:00	Paid 15/04/2019	Change
Cash	14.060	21.100	+7.040
Dividends payable (current liability)	0	7.040	+7.040
Reserves and owners equity	19.560	19.560	0
Total assets = total liabilities and equity	38.270	45.310	+7.040

The decision already reduces the profit kept in the company, so equity does not change; only the moment the money leaves changes.

KEY RESULTS

- ROA = 63,18%; ROE = 92,02% (extension).
- Paid on 15/04/2019: cash +7.040 (21.100); new current liability "dividends payable" 7.040; totals 45.310 m.u.; owners equity unchanged.

Part (d) · 0.75 points

Effect of eliminating the 8% discount to the cooperative; which case gives more profit and which more cash?

Step 1. New sales of the cooperative

$$75 \times 0{,}90 = 67{,}5 \text{ u./month}, \qquad 67{,}5 \times 25 = 1.687{,}5 \text{ m.u./month}, \qquad 12 \times 1.687{,}5 = 20.250 \text{ m.u.}$$

$$\Delta\text{Revenue} = 20.250 - 20.700 = -450 \text{ m.u.} \;\Rightarrow\; \text{Revenue} = 87.750 \text{ m.u.}$$

Step 2. Raw material (hypothesis 4)

In the base case 4.500 tons were consumed for 3.600 units: $4.500/3.600 = 1{,}25$ t/unit $= 2{,}50$ m.u./unit. Selling $12 \times 7{,}5 = 90$ units fewer:

$$\Delta\text{Consumption} = -90 \times 2{,}50 = -225 \text{ m.u.} \;\Rightarrow\; \text{final stock} = 1.800 + 225 = 2.025 \text{ m.u.}$$

Step 3. New results

$$\text{Profit before taxes} = 24.000 - 450 + 225 = 23.775, \qquad \text{Tax} = 0{,}25 \times 23.775 = 5.943{,}75$$

$$\text{Net profit} = 23.775 - 5.943{,}75 = 17.831{,}25, \qquad \text{Remaining} = 17.431{,}25$$

$$\text{Reserves} = 0{,}60 \times 17.431{,}25 = 10.458{,}75, \qquad \text{Dividends} = 0{,}40 \times 17.431{,}25 = 6.972{,}50$$

Step 4. New cash

The cooperative now pays at 60 days: January to October collected, $10 \times 1.687{,}5 = 16.875$ instead of 20.700 (–3.825). The dividend paid falls by 67,50; the tax is not paid in 2018.

$$\text{Cash} = 14.060 - 3.825 + 67{,}50 = 10.302{,}50 \text{ m.u.}$$

Step 5. Changes in the statements

Item	Base	Without discount	Change
Revenue	88.200	87.750	–450
Raw material consumed	9.000	8.775	–225
Net profit	18.000	17.831,25	–168,75
Inventory	1.800	2.025	+225
Accounts receivable	11.450	14.825	+3.375
Cash	14.060	10.302,50	–3.757,50
Total assets	38.270	38.112,50	–157,50

Item	Base	Without discount	Change
Reserves	10.560	10.458,75	-101,25
Unpaid taxes	6.000	5.943,75	-56,25
Total liabilities and equity	38.270	38.112,50	-157,50

Discussion. The **profit is higher with the discount** (18.000 against 17.831,25): removing the discount gains 2 m.u. on each of the 810 units the cooperative still buys ($810 \times 2 = 1.620$ m.u.), but the 90 units it stops buying carried a margin over raw material of $23 - 2{,}50 = 20{,}50$ m.u. each ($90 \times 20{,}50 = 1.845$ m.u.); the net effect before taxes is $1.620 - 1.845 = -225$ m.u. The **cash is also higher with the discount** (14.060 against 10.302,50): the cooperative pays in the same month, while under the general condition November and December stay as receivables. Keeping the discount wins on both counts.

KEY RESULTS

- Cooperative: 67,5 u./month at 25 m.u. = 1.687,5 m.u./month; revenue 87.750 (–450).
- Raw material consumed –225 (90 fewer units × 2,50); net profit 17.831,25 (–168,75).
- Receivables +3.375; inventory +225; cash 10.302,50 (–3.757,50); total assets 38.112,50.
- Profit and cash are both higher with the discount.

Interpretation and checks

- **Dimensional check:** units × m.u./unit and tons × m.u./ton give m.u.; m.u./year × months/12 gives m.u.; ratios are dimensionless.
- **Why ROA and ROE are so high:** the company's annual sales (88.200 m.u.) are ten times its 2017 assets; the ratios are measured on the year-end balance, which already includes most of the year's profit.
- **Accrual vs cash:** net profit 18.000 m.u., but cash grew by only 13.120 m.u. because 11.250 m.u. of sales are uncollected, although many expenses (tax, software, raw material, renting) are still unpaid.
- **Typical mistakes:** depreciating the vehicles for 12 months, forgetting the residual values, treating the issue premium as capital, and putting the whole short-term credit or the 200 m.u. deferred receivable outside current items.